

GLA UNIVERSITY, MATHURA



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JOB BRIDGE – JOB PORTAL **(MERN STACK)**

Submitted By

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Submitted to:

Faculty Name: Akash Kumar Choudhary

Technical Trainer: Shival Pal

DECLARATION

We hereby declare that the mini project titled “**JobBridge** – Job Portal (MERN Stack)” is being developed by us as part of our academic mini project.

We, Vikash Sisodia (Roll No. 2415001774) and Vichitra Solanki (Roll No. 2415001767), hereby confirm that we will work together on the design and development of this project. The work presented in this project will be carried out by us under the guidance of Shival Pal, Technical Trainer.

We further declare that the project work will be our original work and will be completed in accordance with the requirements and guidelines provided by the university.

Team Members:

1. Vikash Sisodia — Roll No. 2415001774
2. Vichitra Solanki — Roll No. 2415001767

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INTRODUCTION :-

About the Project:

JobBridge – Job Portal (MERN Stack) is a full-stack web application designed to connect job seekers with employers through an efficient online platform. The system allows employers to create and manage job listings, while job seekers can browse available jobs, view job details, and submit applications for suitable positions.

The application is developed using the **MERN Stack**, consisting of MongoDB, Express.js, React.js, and Node.js. React.js is used to develop the interactive user interface, while Node.js and Express.js handle the server-side logic and RESTful APIs. MongoDB is used to store and manage application data such as user profiles, job listings, and job applications.

The main purpose of JobBridge is to provide a simple, organized, and user-friendly platform for the recruitment process. It reduces the dependency on traditional job-search methods and provides a centralized system where employers can manage job opportunities and applicants can easily find and apply for relevant jobs.

The project also follows REST API design, clean coding practices, and Git/GitHub version control to ensure that the application is maintainable, scalable, and suitable for collaborative development.

Primary Reason to Choose This Project

We chose to work on **JobBridge – Job Portal (MERN Stack)** because it is a practical and meaningful real-world project that addresses the growing need for a centralized platform to connect job seekers with employers. This project allows us to apply and strengthen our skills in full-stack web development using modern technologies like MongoDB, Express.js, React.js, and Node.js.

Some key reasons for choosing this project are:

- **Real-World Relevance:** Job portals are widely used and have a significant impact on the recruitment process.
- **Skill Enhancement:** It helps us learn and implement full-stack development concepts in a practical way.
- **Problem Solving:** It gives us the opportunity to solve real-life problems related to job searching and hiring.
- **Career-Oriented:** The project is aligned with current industry trends and improves our understanding of web application development, which is valuable for our future careers.
- **Team Collaboration:** It allows us to work as a team, manage tasks, and experience the complete software development lifecycle.

The Main Objective of the Project

The main objective of JobBridge – Job Portal (MERN Stack) is to develop a user-friendly and efficient platform that connects job seekers with employers and simplifies the recruitment process.

The specific objectives are:

- **Provide a centralized platform** for posting and searching job opportunities.
- **Allow job seekers** to easily browse, view, and apply for relevant jobs.
- **Enable employers** to create, manage, and track job listings and applications.
- **Ensure a smooth and secure** experience for all users.
- **Implement** the project using modern web technologies (MERN Stack) to build a scalable and maintainable application.
- **Gain hands-on experience** in full-stack development and apply theoretical knowledge to a real-world project.

Scope Of the Project

The scope of **JobBridge – Job Portal (MERN Stack)** covers the development of a web-based platform that connects job seekers with employers. The system will provide the following functionalities:

- Job seekers can register, create their profile, search and browse available job listings.
- Job seekers can view complete job details and apply for suitable positions.
- Employers can register and manage their profiles.
- Employers can create, update, and manage job listings.
- Employers can view and manage applications received for their posted jobs.
- The system will maintain job seeker, employer, job listing, and application data using MongoDB.
- RESTful APIs will be used for communication between the frontend and backend.
- The application will be designed to be user-friendly, maintainable, and scalable.

Working Methodology Of the Project:

The project will follow a structured full-stack development methodology. The major steps are:

1. **Requirement Analysis:** Identify the requirements of job seekers and employers and define the major system functionalities.
2. **System Design:** Design the application architecture, database structure, user flow, and REST API structure.
3. **Frontend Development:** Develop the user interface using React.js for job searching, job posting, applications, and profile management.
4. **Backend Development:** Develop the server-side application using Node.js and Express.js and implement RESTful APIs.
5. **Database Integration:** Use MongoDB to store and manage users, jobs, and applications.
6. **Testing:** Test different modules and APIs to identify and fix functional or integration issues.
7. **Version Control:** Use Git and GitHub for source-code management and team collaboration.
8. **Deployment:** Integrate the frontend, backend, and database to make the completed application available for use.

Details About the Hardware and the Software

System Requirements: -

Supported Operating System: -

The application can be developed and run on the following operating systems:

- Windows 10 / 11 (64-bit)
- macOS (latest version)
- Linux (Ubuntu 20.04 or above)

Software Required: -

The following software and tools are required for the development and execution of the project:

- Visual Studio Code (Code Editor)
- Node.js (with npm)
- MongoDB (Local installation or MongoDB Atlas)
- Git (Version Control)
- Web Browser (Google Chrome / Mozilla Firefox / Microsoft Edge)

Hardware Requirements: -

The minimum hardware configuration required for smooth development and execution is:

- **Processor** : Intel Core i3 / AMD Ryzen 3 or above
- **RAM** : 8 GB (minimum), 16 GB (recommended)
- **Storage** : 10 – 20 GB of free space
- **Internet** : Stable internet connection (required)
- **Display** : 1366 × 768 resolution or above

Listing Out testing technology

The following tools and methods are used for testing the application:

- Manual Testing (Functional and UI Testing)
- Postman (for testing REST APIs)
- Browser Testing (Chrome, Firefox, Edge)
- MongoDB Data Verification (to check stored data)
- Console Logging and Error Debugging (using VS Code Developer Tools)

Frontend and Backend: -

Frontend:

- React.js (with Vite)
- HTML, CSS, JavaScript
- Optional: Tailwind CSS (for styling)

Backend:

- Node.js
- Express.js
- RESTful APIs

Database:

- MongoDB (Local or MongoDB Atlas)

Module Description:

The **JobBridge – Job Portal (MERN Stack)** is divided into several modules, each responsible for a specific set of functionalities. The modules work together to provide a complete and user-friendly platform for job seekers and employers.

1. User Authentication Module

This module handles user registration, login, and authentication for both job seekers and employers. It ensures secure access to the system using JWT (JSON Web Tokens) and password hashing.

2. Job Seeker Module

This module allows job seekers to create and manage their profiles, search and browse job listings, view job details, and apply for suitable positions. Job seekers can also track the status of their applications.

3. Employer Module

This module enables employers to register and manage their company profiles. Employers can create, update, and manage job postings, and view and manage applications received for their jobs.

4. Job Listing Module

This module handles the creation, storage, and display of job listings. It provides features like search, filter, and sort to help job seekers find relevant jobs easily.

5. Application Management Module

This module manages job applications. Job seekers can submit and track their applications, while employers can view, shortlist, and manage the applications received for each job posting.

6. Database Management Module

This module is responsible for storing and managing all application data such as user information, job listings, and job applications using MongoDB.

7. REST API and Backend Module

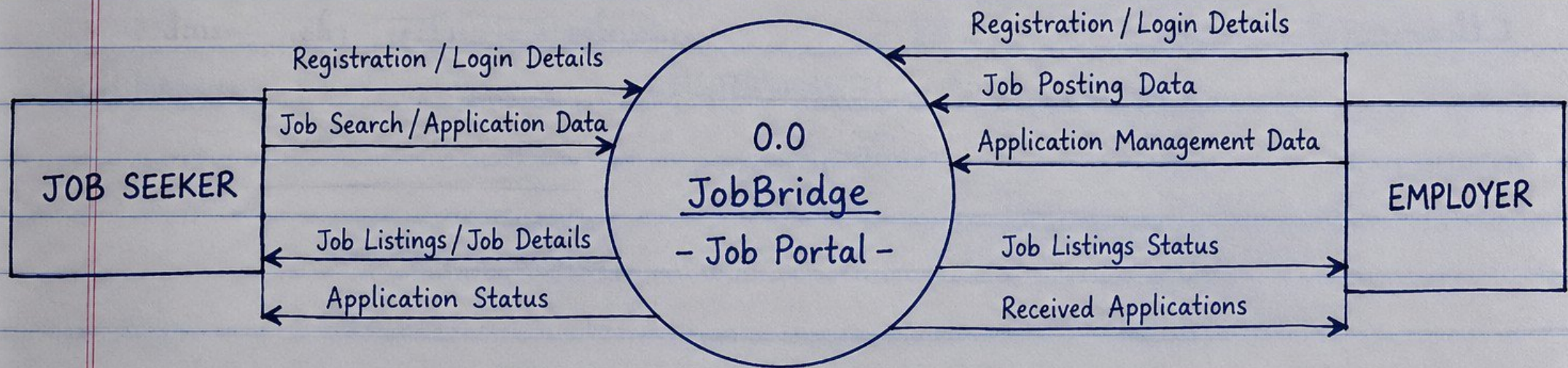
This module is developed using Node.js and Express.js. It handles the server-side logic, defines RESTful APIs, and manages communication between the frontend and database.

8. Frontend Module

This module is developed using React.js. It provides a responsive and user-friendly interface for both job seekers and employers, including pages for authentication, job search, job management, and application tracking.

Data Flow Diagrams

DFD - LEVEL 0 (CONTEXT DIAGRAM)



Note:-

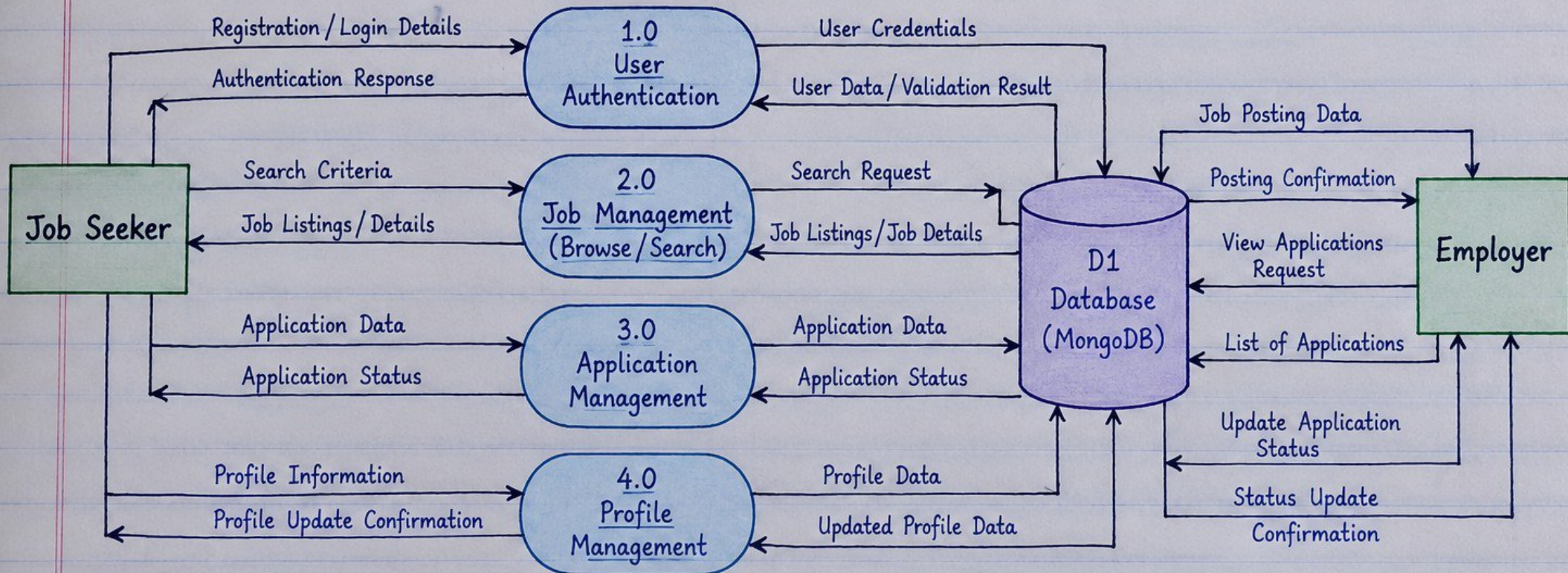
- 1) Level 0 DFD (Context Diagram) shows the overall system as a single process.
- 2) It shows the external entities (Job Seeker and Employer) and the major data flows between them and the system.

Data Flow Diagrams

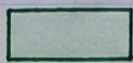
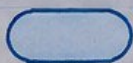
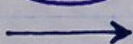
DFD - Level 1

(JobBridge - Job Portal)

Level 1 DFD shows the major processes of the system, how external entities interact with these processes and how data flows between processes and the database.



Legend:-

-  : External Entity
-  : Process
-  : Data Store (Database)
-  : Data Flow

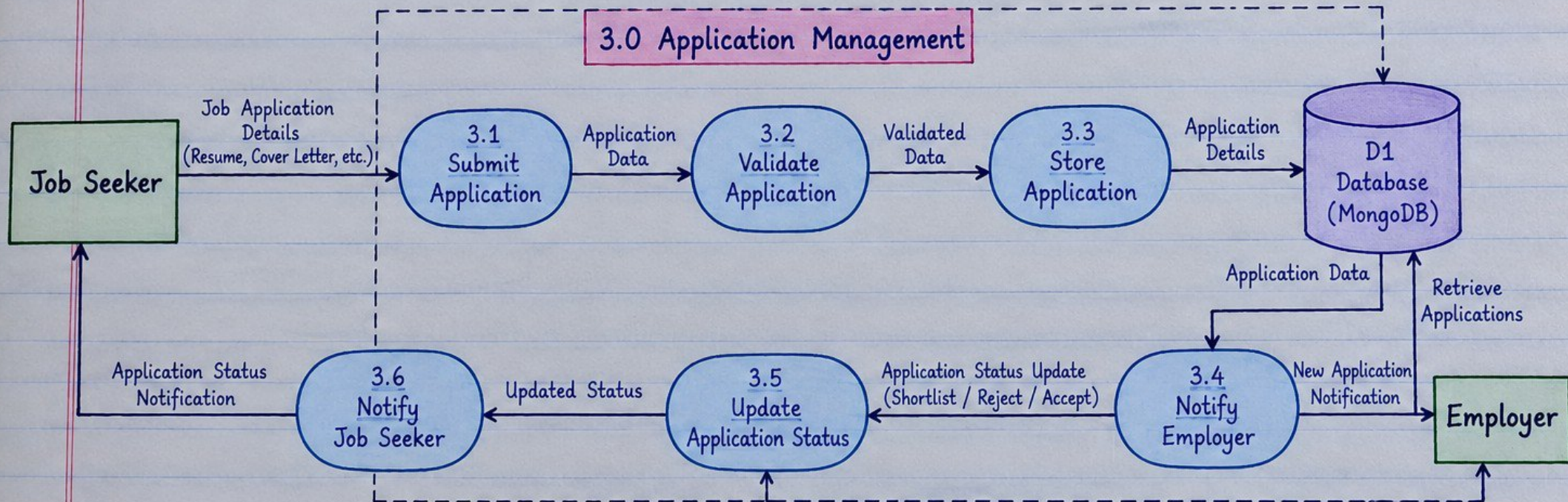
Note:-

- 1) Level 1 DFD breaks the main system into major processes.
- 2) It shows how Job Seeker and Employer interact with different processes.
- 3) The database (MongoDB) stores and provides data to all processes.

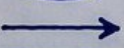
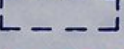
Data Flow Diagrams

DFD - Level 2 (Application Management Process)

Level 2 DFD shows the detailed working of a specific process from Level 1. Here, we have broken down the "Application Management" process into smaller sub-processes.



Legend:-

-  : External Entity
-  : Process
-  : Data Store (Database)
-  : Data Flow
-  : Process Boundary

Note:-

- 1) This is Level 2 DFD for the "Application Management" process (3.0) from Level 1.
- 2) It shows the detailed steps involved in handling a job application.
- 3) Data is stored in MongoDB and both Job Seeker and Employer are notified about the application status.
- 4) Similarly, Level 2 DFD can be created for other major processes if required.

References:

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2. Node.js Documentation – Official Node.js documentation.
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3. Express.js Documentation – Official Express.js documentation.
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